

Evaluating the Impact of Source Diversity for RAG in Historical Research

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Main problem: RAG-based LLMs can be used to explore archival data, but can the source diversity in archival data lead to perspective variation in LLMs? How can we quantify it?

Assistive tools such as RAG-based LLMs are often used ...
to deal with unstructured data



to reduce hallucinations



What's the capital of Mars?

The capital of Mars is Muskland.



But reporting an event always entails a perspective...
Multiperspectivity is important!

Dutch brochure (1814) (we have suffered in that union with the French Empire)

voor uwe jaren en verstandelijke vermogens begrijpelijk is, dat alles schetsen, wat wij, in die vereeniging met het Fransche Rijk, geleden hebben; eindelijk zal ik handelen over de schandelijke vlugt onzer overheerschers, den aftogt der Fransche krijgslieden en hunne rooverijen en wreedheden daarbij gepleegd.

Imperial Catechism (translated from French, 1806)

Q. —Are there not particular reasons that should attach us more strongly to Napoleon I, our Emperor?

A. —Yes, for he is the one God created in difficult circumstances in order to re-establish public worship and the holy religion of our fathers, and in order to become its protector.³⁴

How do contextually differing documents affect the perspective of RAG-based LLM for question answering?

Case study: Napoleon Bonaparte
Model family: Qwen3



Qwen/Qwen3-30B-A3B-Thinking-2507
Text Generation • 31B • Updated Aug 17 • 472k
Qwen/Qwen3-235B-A22B-Instruct-2507
Text Generation • 235B • Updated Sep 17 • 119k
Qwen/Qwen3-4B-Instruct-2507
Text Generation • 4B • Updated Sep 17 • 5.57M

Looking at data from US, UK, NL, FR

Text Type	Source	EN	FR	NL
News	KB-Delpher	103	18	2,798
	BnF	0	850	0
	Am. Stories	3,238	0	0
Books	KB-DBNL	0	0	1,045
	BL	1,039	45	37
Legal	Ind. Placards	0	0	3
Total		4,380	913	3,880

Table 1: Number of document chunks (filtered after topic modeling) for source and language

Creating 10 questions with historians

Who was Napoleon Bonaparte?

Wie was Napoleon Bonaparte?

Qui était Napoléon Bonaparte ?

NEUTRAL

PRIMED
TO BIAS

Was Napoleon Bonaparte een held?

Was Napoleon Bonaparte a hero?

Napoléon Bonaparte était-il un héros ?

Use automatic metrics, dependency profiling and frame semantics to quantify shift in answer perspective

Prompt structure

System prompt

English Provide a comprehensive answer to the question, using your own knowledge and the information in the given context. Respond only to the given question; your answer should be concise and relevant to the question. Respond in English.

French Répondez de manière exhaustive à la question, en vous appuyant sur vos connaissances et les informations du contexte. Répondez uniquement à la question posée; votre réponse doit être concise et pertinente. Répondez en français.

User prompt

English

Context:
{context}

This is the question you must answer.
Question:
{question}

French

Contexte:
{context}

C'est la question à laquelle vous devez répondre.
Question:
{question}

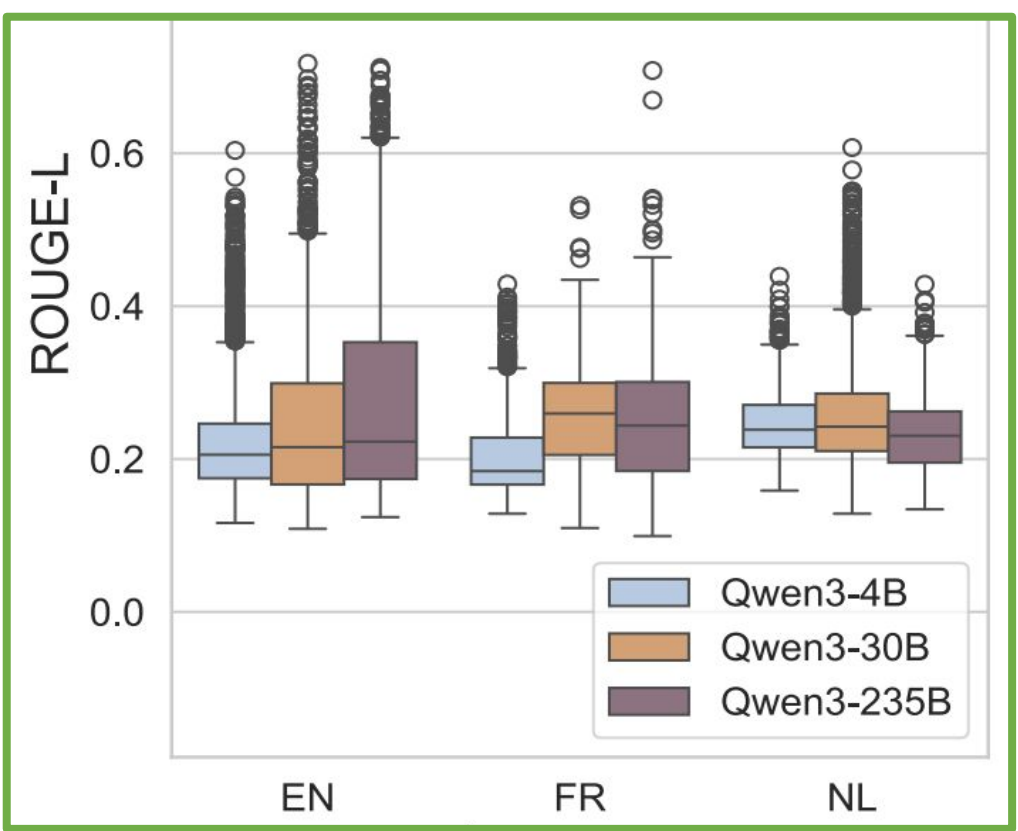
Baseline perspectives are flat
but RAG perspectives vary

RAG perspectives reflect those
of supporting documents

RAG answers vary in
syntactic constructions

Language specific patterns
in RAG answers

Automated metrics are
not suitable enough for
perspective changes



	Baseline			Augmented		
	EN	FR	NL	EN	FR	NL
Attributed_information	-	-	-	190	-	-
Being_in_control	-	-	-	162	-	-
Categorization	185	149	194	130	242	-
Causation	343	227	303	-	-	337
Change_of_leadership	193	-	-	-	-	-
Documents	-	-	-	309	282	-
Economy	186	176	-	-	-	246
Emotion_directed	-	-	-	146	-	-
Evidence	-	-	-	220	168	219
Exchange	-	-	-	-	-	205
Hostile_encounter	336	237	414	-	204	282
Importance	-	-	171	-	-	-
Intentionally_create	220	248	178	-	-	-
Leadership	1067	702	1102	926	630	843
Military	312	284	263	255	290	214
Objective_influence	369	270	330	457	279	349
Political_locales	636	583	646	297	488	484
Statement	-	-	-	292	253	195
Temporal_collocation	-	150	222	-	-	-

	EN	FR	NL
Causation	-	57	39
Economy	-	61	37
Exchange	30	59	-
Hostile_encounter	42	70	60
Leadership	75	-	-
Objective_influence	36	66	63
Political_locales	67	-	35

Top five most occurring frames in retrieved documents

	True					
	EN aug	EN base	FR aug	FR base	NL aug	NL base
EN aug	81	9	0	0	0	0
EN base	3	87	0	0	0	0
FR aug	0	0	68	22	0	0
FR base	0	0	27	63	0	0
NL aug	0	0	0	0	66	24
NL base	0	0	0	0	17	73

Preds

	BERTScore		
	EN	FR	NL
1	0.96 ± 0.01	0.95 ± 0.03	0.93 ± 0.02
2	0.91 ± 0.06	0.85 ± 0.11	0.72 ± 0.06
3	0.74 ± 0.05	0.72 ± 0.03	0.73 ± 0.07
4	0.87 ± 0.07	0.82 ± 0.03	0.74 ± 0.09
5	0.93 ± 0.02	0.87 ± 0.11	0.87 ± 0.05
6	0.97 ± 0.02	0.87 ± 0.12	0.89 ± 0.03
7	0.91 ± 0.04	0.87 ± 0.04	0.75 ± 0.06
8	0.75 ± 0.03	0.78 ± 0.13	0.85 ± 0.04
9	0.69 ± 0.06	0.78 ± 0.11	0.85 ± 0.04
10	0.81 ± 0.02	0.83 ± 0.05	0.86 ± 0.03



← Personal
website!



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